

Identifying Pacific Northwest Birds from Acoustic Recordings

Identifying Pacific Northwest Birds from Acoustic Recordings: A Deep Learning Approach

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Abstract

This report presents a convolutional neural network (CNN) pipeline for classifying 12 Pacific Northwest bird species from mel spectrogram representations of 3-second audio clips. A total of 1,981 recordings from the Xeno-Canto archive are stored as 128×517 mel spectrograms. Two binary CNN architectures distinguish Bewick's Wren from Dark-eyed Junco: Architecture A achieves 61.0% test accuracy with a strong Wren bias, while Architecture B with Dropout achieves 73.2% with more balanced recall. Three 12-class CNNs trained from scratch perform near the 8.3% chance level (best: 9.4%), failing to learn species-specific features from the limited dataset. An EfficientNet-BO backbone pretrained on ImageNet and fine-tuned in two phases achieves 67.2% test accuracy with macro F1 of 0.53, confirming that transfer learning is essential at this dataset scale. Two of three unlabeled test clips are flagged as containing multiple species.

1. Introduction

Automated species identification from acoustic field recordings is a valuable application of machine learning in ecology and conservation biology. Manual review of large audio archives requires specialist expertise and is prohibitively time-consuming at scale, making computational approaches essential for programs such as BirdNET (Kahl et al., 2021).

Bird vocalizations are well suited to image-based deep learning because they can be represented as mel spectrograms: 2D time-frequency images where pixel intensity encodes acoustic energy on a perceptually motivated frequency scale. CNNs exploit the local spatial structure of these images to learn call-specific feature detectors – the same inductive bias that makes them effective for natural image classification (James et al., 2021).

This report covers a binary classification task (Bewick's Wren vs. Dark-eyed Junco), a 12-class from-scratch CNN comparison, EfficientNet-BO transfer learning, and sliding-window analysis of

three unlabeled test clips. Code and figures are at: <https://github.com/gpskentzos/DATA5322-BirdSongs-of-Seattle>

2. Background

Mel spectrograms are computed by applying a Short-Time Fourier Transform to overlapping audio frames, mapping the power spectrum through a mel filterbank (approximately linear below 1 kHz, logarithmic above), and converting to decibels. All recordings use librosa defaults: 22,050 Hz sample rate, 128 mel bins, hop size 128, window 512, yielding 128×517 arrays in the range $[-80, 0]$ dB (McFee et al., 2015). Values are normalized to $[0, 1]$ via $(x + 80)/80$ before training.

CNNs learn local feature detectors that scan the full spectrogram via weight sharing, drastically reducing parameters relative to a fully connected approach on the 66,176-dimensional input. Global Average Pooling (GAP) aggregates the final feature maps to a fixed-length vector without flattening, further controlling model size. Dropout (Srivastava et al., 2014) and Batch Normalization (Ioffe and Szegedy, 2015) are evaluated as regularization strategies. All models are trained with the Adam optimizer (Kingma and Ba, 2015), EarlyStopping (patience = 12), and ReduceLROnPlateau (factor = 0.5, patience = 5).

The dataset spans a 17:1 imbalance from House Sparrow (630 recordings) to Northern Flicker (37). Inverse-frequency class weighting is applied during multi-class training:

$$w_c = \frac{N}{C \cdot N_c}$$

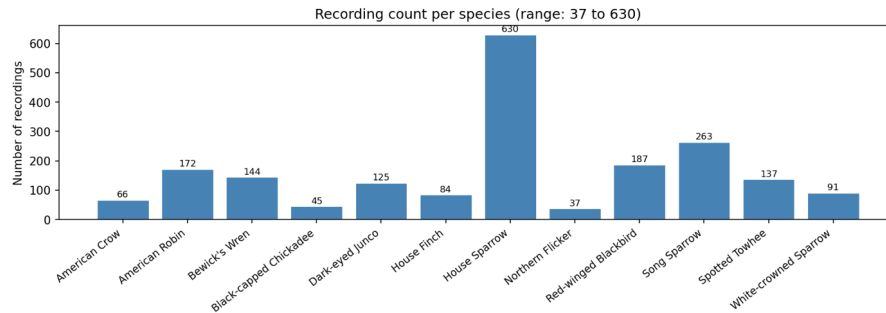
where N is total training samples, $C = 12$ classes, and N_c is the count for class c . Weights range from 0.261 (House Sparrow) to 4.610 (Northern Flicker).

3. Methodology

Dataset. 1,981 recordings across 12 Pacific Northwest species, sourced from Xeno-Canto (Vellinga and Planque, 2015) and filtered by the instructor for quality and duration. Data are stored in HDF5 format as $(128, 517, N)$ arrays per species. Table 1 lists the species and recording counts.

Table 1: Species and recording counts

Code	Common Name	N	Code	Common Name	N
amecro	American Crow	66	houspa	House Sparrow	630
amerob	American Robin	172	norfli	Northern Flicker	37
bewwre	Bewick's Wren	144	rewbla	Red-winged Blackbird	187
bkcchi	Black-capped Chickadee	45	sonspa	Song Sparrow	263
daejun	Dark-eyed Junco	125	spotow	Spotted Towhee	137
houfin	House Finch	84	whcspa	White-crowned Sparrow	91



Class distribution bar chart

Figure 1: Recording counts per species. The 17:1 imbalance ratio between House Sparrow and Northern Flicker requires class weighting during training.

Splits. Stratified 70/15/15 split: 1,383 training, 299 validation, 299 test samples.

Binary task. Bewick's Wren (144) vs. Dark-eyed Junco (125). Two architectures: Architecture A (3 conv blocks + GAP + Dense(64), no regularization, 100,993 params) and Architecture B (same backbone + Dropout(0.25) between blocks + Dropout(0.40) in head, 109,313 params).

Multi-class task. All 12 species with class weighting. Three architectures share the same 3-conv backbone: (1) Baseline CNN, (2) CNN + Dropout, (3) CNN + BatchNorm.

Transfer learning. EfficientNet-Bo (Tan and Le, 2019) pretrained on ImageNet. Single-channel spectrograms are replicated to 3 channels and rescaled to [0, 255] for compatibility with ImageNet weights. Phase 1: backbone frozen, head-only training at $lr = 1e-3$. Phase 2: top 40 of 238 backbone layers unfrozen, full network at $lr = 1e-4$.

Test-clip analysis. Three unlabeled MP3 clips (23.3 s, 5.3 s, 15.9 s) processed with a 3-second sliding window (1-second stride). Each window is classified by the best model. A second species is flagged if its mean window probability exceeds 0.20.

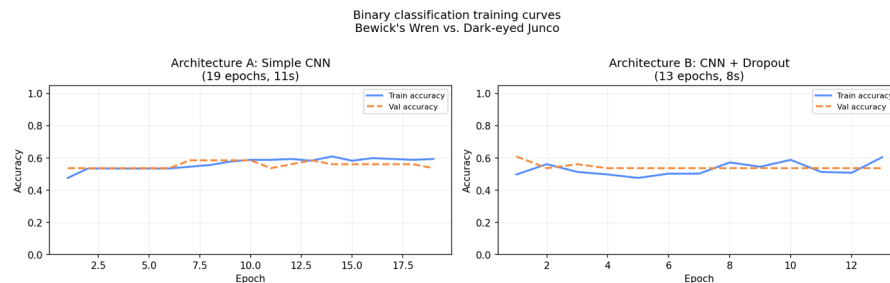
4. Results

4.1 Binary Classification

Table 2: Binary architecture comparison

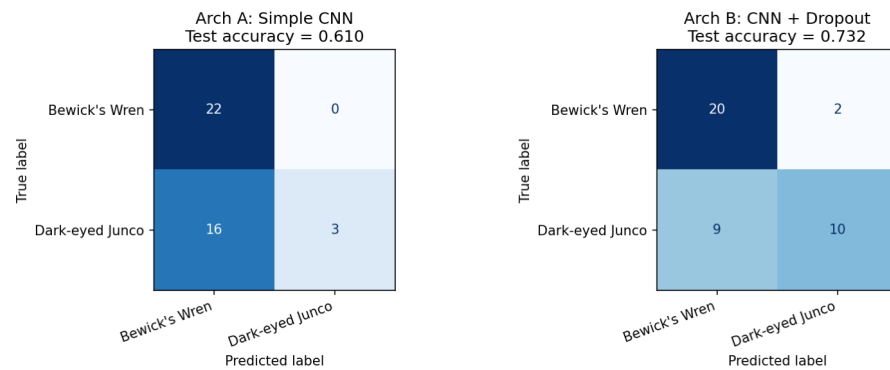
Architecture	Params	Epochs	Time	Val Acc	Test Acc
A: Simple CNN	100,993	19	11 s	0.585	0.610
B: CNN + Dropout	109,313	13	8 s	0.610	0.732

Architecture A achieves 61.0% but exhibits a strong Wren bias (recall = 1.00 for Wren, 0.16 for Junco) – it defaults to Bewick's Wren when uncertain. Architecture B achieves 73.2% with more balanced performance (recall = 0.91 for Wren, 0.53 for Junco); Dropout regularization prevented the model from collapsing to the majority class.



Binary training curves

Figure 2: Training and validation accuracy for both binary architectures. Architecture B achieves higher validation accuracy (0.610) and converges more stably.



Binary confusion matrices

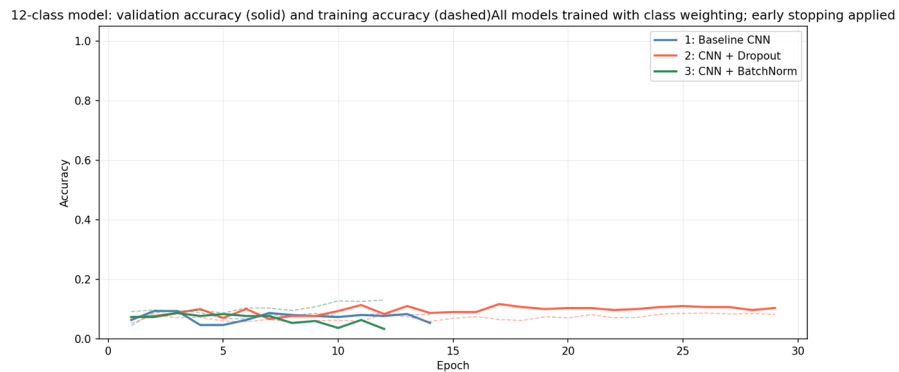
Figure 3: Confusion matrices (41 test samples). Architecture A classifies all 22 Wren recordings correctly but misses 16 of 19 Junco. Architecture B correctly classifies 20 of 22 Wren and 10 of 19 Junco.

4.2 Multi-class Classification

Table 3: Multi-class architecture comparison

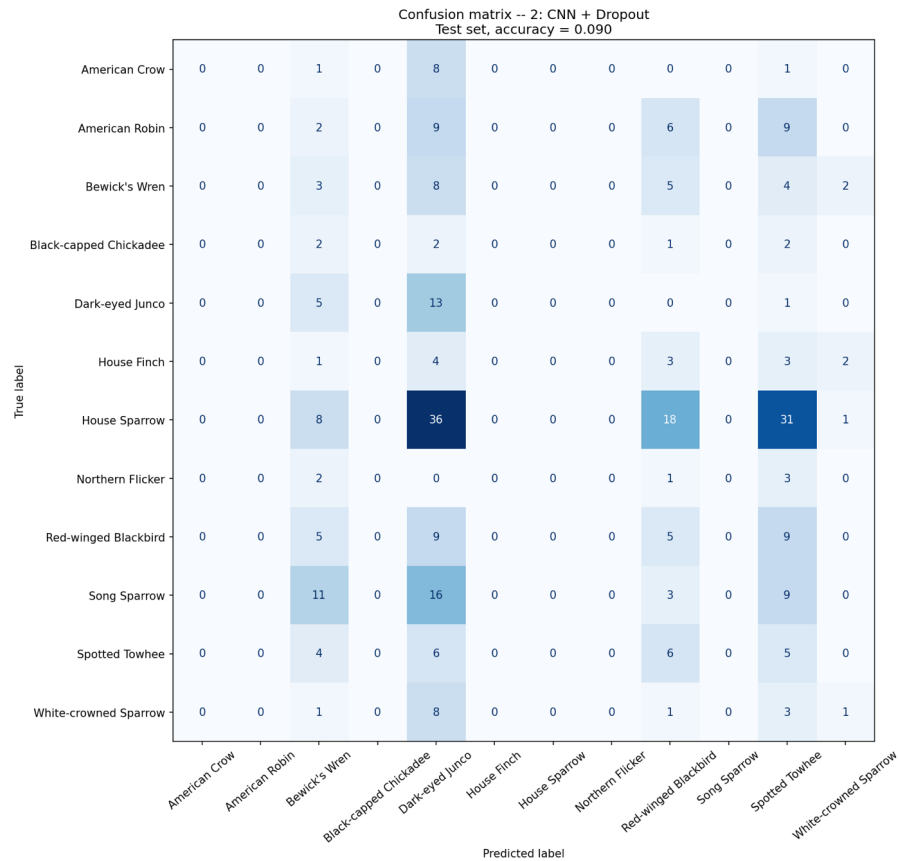
Architecture	Params	Epochs	Time	Val Acc	Test Acc
1: Baseline CNN	110,732	14	56 s	0.094	0.094
2: CNN + Dropout	128,780	29	122 s	0.117	0.090
3: CNN + BatchNorm	129,676	12	69 s	0.087	0.054

All three architectures perform near the 8.3% chance level. CNN+Dropout (best by validation accuracy) shows slight bias toward Dark-eyed Junco (recall = 0.68) while 8 of 12 species have zero recall. The class weighting reduced majority-class collapse but was insufficient to produce generalizable features from approximately 115 training samples per class.



Multi-class training curves

Figure 4: Validation accuracy for three multi-class architectures. All plateau below 12%, indicating failure to learn meaningful species representations from scratch.



Multi-class confusion matrix

Figure 5: Confusion matrix for the best from-scratch model (CNN+Dropout, 9.0% test accuracy).

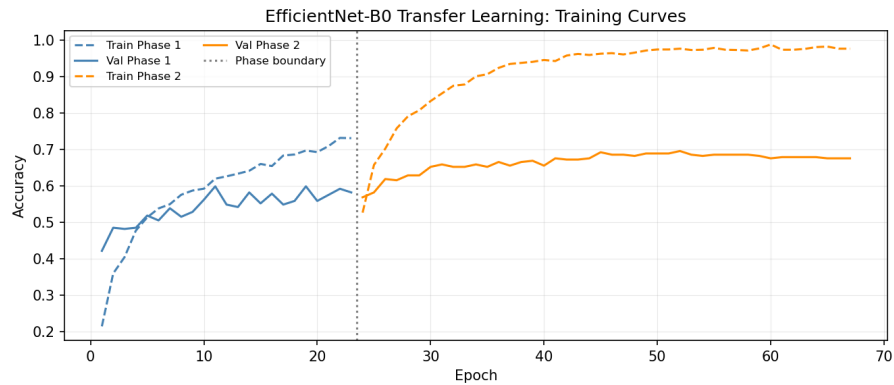
4.3 Transfer Learning: EfficientNet-Bo

Table 4: Full model comparison (12-class test set)

Model	Params	Test Acc	Macro F1	Phase 1	Phase 2
MC Baseline CNN	110,732	0.094	0.014	—	—
MC Dropout CNN	128,780	0.090	0.051	—	—
MC BatchNorm CNN	129,676	0.054	0.027	—	—
EfficientNet-Bo	4,380,591	0.672	0.533	67.1 s	247.3 s

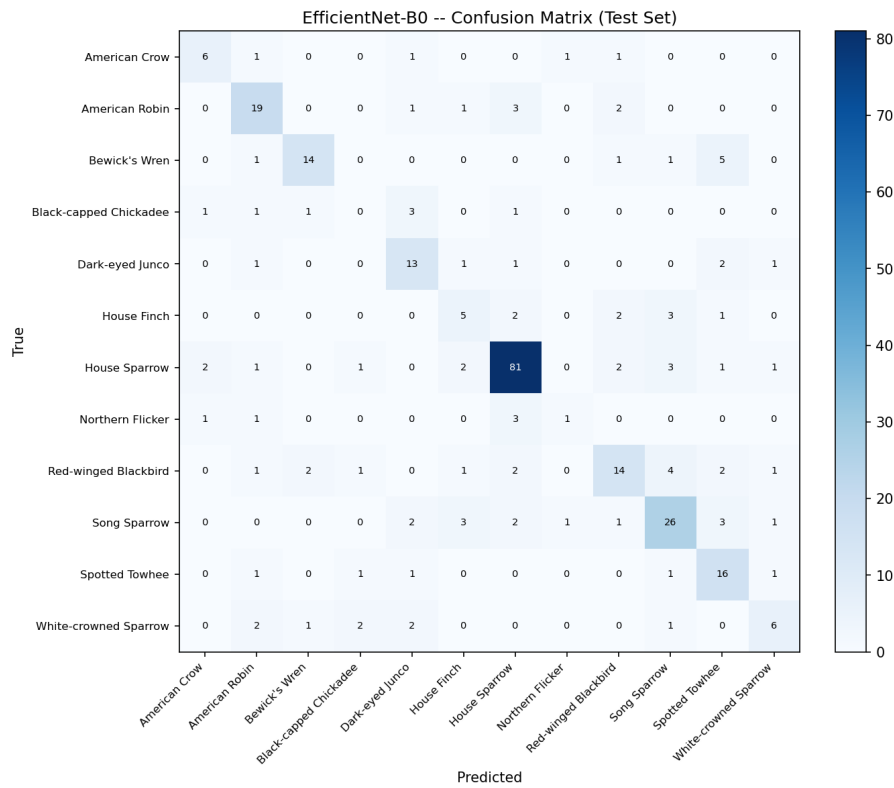
EfficientNet achieves 67.2% test accuracy and macro F1 of 0.53 – roughly 10× the macro F1 of the best from-scratch model. Phase 2 fine-tuning showed rapid overfitting (training accuracy ~93%, validation ~67%), consistent with the small dataset limiting further backbone adaptation. Non-zero recall is achieved for 11 of 12 species; the sole failure is Black-capped Chickadee (31 training,

7 test samples), where support is too small for learning or reliable evaluation.



EfficientNet training curves

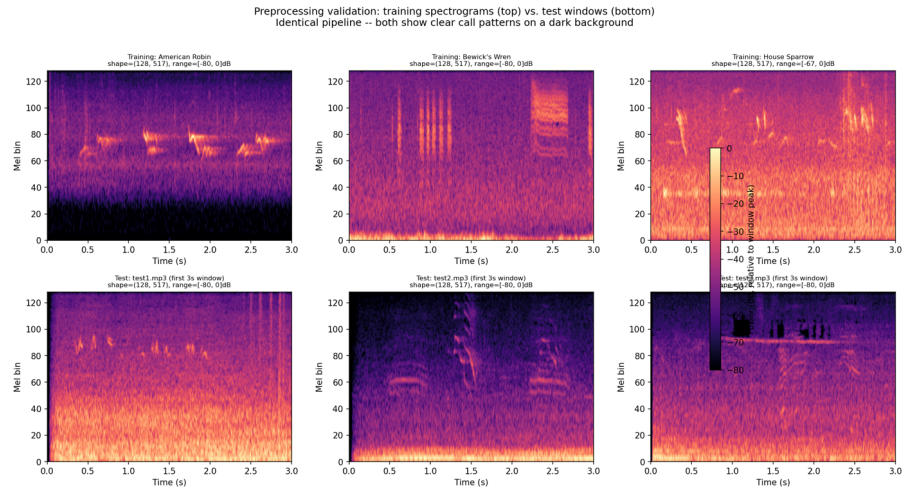
Figure 7: EfficientNet-Bo training curves. Phase 1 (blue) shows steady head adaptation. Phase 2 (orange) shows rapid training accuracy gain with limited validation improvement, indicating overfitting.



EfficientNet confusion matrix

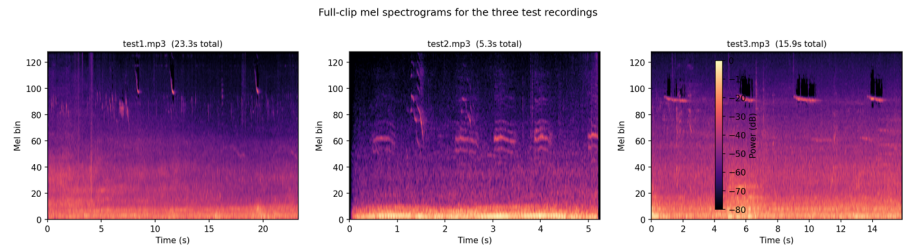
Figure 8: EfficientNet-Bo confusion matrix. Unlike the from-scratch models, predictions are distributed across all 12 species with a bright diagonal, indicating genuine multi-class discrimination.

4.4 External Test-Clip Predictions



Preprocessing validation

Figure 9: Preprocessing validation. Training spectrograms (top row) and first windows from each test clip (bottom row) share identical shape and dB range, confirming the test pipeline is correct.



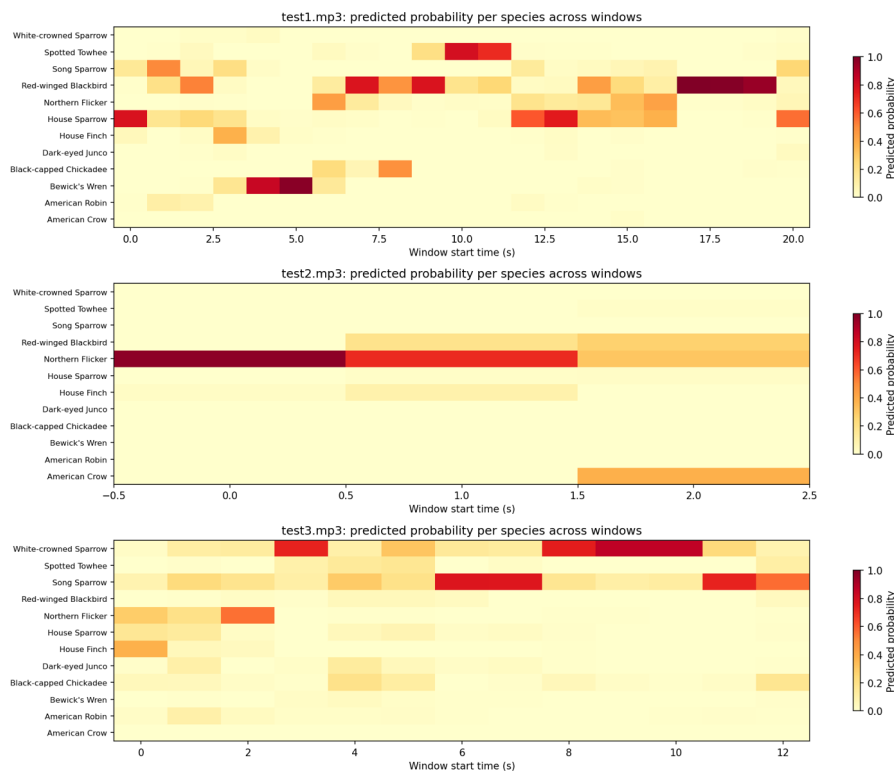
Full-clip spectrograms

Figure 10: Full mel spectrograms for the three test clips. Test1.mp3 and test3.mp3 show multiple distinct call events across their duration, consistent with varied vocalizations or multiple birds.

Table 5: Test-clip predictions (EfficientNet-Bo, 67.2% test accuracy)

Clip	Duration	Windows	Top species	Confidence	2nd species	2nd conf.	Multiple birds?
test1.mp3	23.3 s	21	Red-winged Blackbird	0.340	House Sparrow	0.212	Yes
test2.mp3	5.3 s	3	Northern Flicker	0.664	Red-winged Blackbird	0.151	No
test3.mp3	15.9 s	13	White-crowned Sparrow	0.343	Song Sparrow	0.331	Yes

Test2.mp3 produces the most confident prediction (Northern Flicker, 66.4%). Test1.mp3 and test3.mp3 both trigger the multi-bird flag: House Sparrow (0.212) accompanies Red-winged Blackbird in test1, and White-crowned Sparrow (0.343) and Song Sparrow (0.331) are nearly tied in test3 – either two sparrow species overlap or a single sparrow produces calls with features of both.



Prediction heatmap

Figure 11: Per-window probability heatmap. Northern Flicker dominates test2.mp3 across all windows. Test3.mp3 shows White-crowned Sparrow and Song Sparrow sharing probability mass across time.

5. Discussion

5.1 Limitations

The primary limitation is dataset size relative to input dimensionality. With 1,383 training samples across 12 classes (average 115 per class, minimum 25 for Northern Flicker) and a 66,176-dimensional spectrogram input, from-scratch CNNs cannot learn generalizable features. A bird call typically occupies 0.5–1 second of a 3-second window, leaving the majority of each spectrogram as background noise at the -80 dB floor – further diluting the gradient signal per sample. EfficientNet succeeds where the custom CNNs fail precisely because its pretrained weights provide strong feature detectors that require only a few examples to redirect toward

spectrogram patterns, rather than learning from noise.

Total training time for all six models was 581 seconds (9.7 minutes) on an Apple M-series GPU via `tensorflow-metal`: the five from-scratch models completed in 267 seconds combined; EfficientNet required 314 seconds (Phase 1: 67 s, Phase 2: 247 s). The full notebook ran in 598 seconds (10 minutes).

5.2 Species Difficulty and Acoustic Characteristics

EfficientNet recall patterns map directly onto acoustic structure. Species with compact, distinctive call shapes – American Crow (sparse low-frequency bursts below 4 kHz), Red-winged Blackbird (narrow ascending whistle from 1–4 kHz), Northern Flicker (evenly spaced low-frequency pulses), and Spotted Towhee (buzzy trills with visible harmonic stack) – achieve F1 scores of 0.55–0.63 because their spectrogram signatures are visually isolated from competing classes. The sparrow group (Song Sparrow F1 0.67, White-crowned Sparrow 0.48, House Sparrow 0.86) all vocalize densely in the 2–6 kHz band with similar melodic phrasing; their spectral overlap drives the `test3.mp3` confusion (White-crowned and Song Sparrow nearly tied at 0.343 vs. 0.331). Black-capped Chickadee (recall 0.00) has an audibly distinctive “fee-bee” whistle, but 31 training examples and 7 test samples are insufficient to learn or evaluate its signature reliably.

5.3 Alternative Models and Why Neural Networks

Several non-neural alternatives could address this classification task. **SVMs with MFCC features** are the traditional baseline for audio classification: 13–40 Mel Frequency Cepstral Coefficient values per frame are summarized by their mean and variance across the clip, producing a compact fixed-length feature vector for an RBF kernel SVM. This approach was state of the art for bird classification before deep learning (Briggs et al., 2012) but requires hand-crafted features that encode domain assumptions and cannot adapt to the full richness of the spectrogram. **Random Forests on MFCC features** offer similar interpretability with multi-class handling and feature importance diagnostics, but share the same limitation of fixed, handcrafted representations. **k-Nearest Neighbors on raw or PCA-compressed spectrograms** requires no training but is prohibitively slow at inference and sensitive to irrelevant background noise in the majority of each spectrogram frame.

Neural networks are the appropriate choice for this task for three reasons. First, the input is a 2D structured image with spatially local, repeating patterns – convolutional weight sharing is specifically designed to exploit this structure. Second, learned feature hierarchies can adapt to the specific acoustic patterns of each species without domain expertise: shallow filters detect harmonic partials and onset transients; deeper filters combine these into species-specific motifs that no hand-crafted MFCC can represent. Third, and most importantly demonstrated here, CNNs support transfer learning: the 10× improvement in macro F1 from EfficientNet (0.53 vs. 0.05) is possible only because convolutional features learned from millions of images transfer to spectrogram classification, enabling strong performance from 115 training samples per class – a

regime where any fixed-feature approach would fail.

6. Conclusions

A from-scratch binary CNN achieves 73.2% accuracy distinguishing Bewick's Wren from Dark-eyed Junco, confirming the preprocessing pipeline and architecture are sound. Three 12-class CNNs trained from scratch fail at near-chance accuracy due to the fundamental mismatch between dataset size (115 samples per class average) and input dimensionality (66,176 features). EfficientNet-Bo fine-tuned from ImageNet weights achieves 67.2% test accuracy and macro F1 of 0.53 – a 10× improvement over the best from-scratch model – establishing that transfer learning is essential for effective bird call classification at this dataset scale. The most impactful future directions are SpecAugment data augmentation (Park et al., 2019) and domain-adapted pretrained models such as BirdNET (Kahl et al., 2021).

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